

Predictive Analysis of Medicine Demand Using Historical and Seasonal Data in the Pharmaceutical Industry

Introduction

Pharmaceutical companies always wrestle with one big question: how do you predict how much medicine people will need? They have to plan production and distribution way ahead of time, but real-life demand never stands still. It swings with the seasons, changes as public health trends shift, and sometimes spikes out of nowhere because of sudden outbreaks.

This study digs into how using old data and seasonal trends can help nail down those demand forecasts. It matters—a lot. When companies get these predictions wrong, shelves run empty or, just as bad, warehouses overflow with unused meds. Either way, patients suffer and businesses take a hit.

A lot of companies still lean on basic forecasting tricks. They often ignore seasonal swings or what's happening in public health. There's also a real gap in how well people understand what modern data analysis can do. Smart forecasting tools can give companies a much clearer view, helping them make sharper decisions.

Sure, some researchers have looked at forecasting in retail and healthcare, using things like time series models and machine learning. But when it comes to pharmaceutical products and how they behave seasonally, there's not much out there. This research steps in to fill that gap.

Problem Statement

Guessing how much medicine people will need isn't easy for drug companies. It's a constant headache. If they guess too high, they end up stuck with shelves full of extra pills nobody wants. That's money wasted. If they guess too low, people can't get their medication on time, and treatments get delayed.

And it's not just about the numbers. The way medicines move through the supply chain makes things even trickier. Sometimes drugs go straight from labs to hospitals or pharmacies. Other times, they pass through wholesalers first. Each route has its own

quirks—different buying habits, timing, and customer needs. Because of all this, the usual forecasting tools just don't cut it. They miss all the weird ups and downs you see in real life.

Drug companies pour time and money into gathering huge piles of data to help with forecasting. But weirdly, a lot of them still rely on simple methods that skip over things like seasonal spikes or sudden changes in health trends. The forecasts end up being off the mark, which doesn't help anyone. There's a real need to dig deeper into the data—look at what's happened before, pay attention to seasonal swings, and build smarter models. That's how companies can actually make better calls and keep the whole supply chain running smoothly.

Research Objectives

Forecasting demand for asthma medications is a tough nut to crack for drug companies. Get it wrong, and everything gets messy—inventory piles up or runs dry, production plans get thrown off, and patients end up frustrated. If you guess too low, you risk empty shelves, treatment delays, and even serious health risks. Overshoot, and you're left with warehouses full of unused stock and money down the drain.

Things get even trickier because the pharma supply chain isn't simple. Some companies ship straight from their labs to hospitals or pharmacies. Others go through wholesalers first. Each path has its own quirks—different timelines, customer habits, and demand patterns. Trying to build a forecasting model that fits all these moving parts isn't easy, especially when your data doesn't capture these real-world complications.

This research zooms in on the asthma medication market and aims to build a forecasting model using only public data. Since we can't tap into Chiesi's internal numbers, we'll stick to sources anyone can access and trust.

Public Data Sources to Be Used

- **Pharmaceutical sales data:**
 - [Datos.gob.es](https://datos.gob.es): Spain's national open data portal, which includes health and pharmaceutical consumption datasets.
 - [IQVIA MIDAS Global Data](https://www.iqvia.com/midas): Offers global pharmaceutical market data, including retail and non-retail channels.
 - [Telotrón®](https://www.telotron.com) and [BPS databases](https://www.bps.com): Real-world clinical data on asthma incidence and prescriptions in Spain. [\[ispor.org\]](https://www.ispor.org)
- **Seasonal and environmental data:**
 - Spanish National Hospital Discharge Database (CMBD): Provides hospital admission records for asthma and other respiratory diseases. [\[repisalud.isciii.es\]](https://repisalud.isciii.es)

- [Eurostat Respiratory Disease Statistics](#): Includes EU-wide data on respiratory-related deaths and hospitalizations. [\[ec.europa.eu\]](#)
- [European Environment Agency \(EEA\)](#): Offers data on air pollution and its impact on asthma and other respiratory conditions. [\[eea.europa.eu\]](#)
- [WHO Air Pollution Data Portal](#): Global data on air quality and its health effects, including asthma. [\[who.int\]](#)

These public datasets matter a lot. They let us study demand patterns without risking company secrets. Plus, they open the door to looking at outside factors—think air pollution, seasonal bugs, or hospital admission rates—and how they push demand up or down. By combining this information, the goal is to build a forecasting model that's not just accurate, but also practical for companies juggling multiple supply routes.

Background Analysis

Asthma sticks around for life. It makes breathing tough because the airways get inflamed. It's one of the most common chronic diseases not just in Europe, but everywhere. In Spain, a lot of people—including kids—deal with asthma every day. Managing it takes constant medication. People depend on inhalers, bronchodilators, and corticosteroids they need to get easily, just to keep living normally.

The market for asthma medication is huge in the pharmaceutical world. There's a wide range of products out there, each with its own supply chain. Some go straight from the lab to hospitals or pharmacies. Others move through wholesalers first. All these different routes make managing inventory and predicting demand complicated. Every supply chain works a little differently—different buying habits, delivery schedules, and customer needs. That makes demand forecasting a real challenge.

Pharma companies need to get their demand forecasts right. They have to plan production, control inventory, and make sure meds are on hand when people need them. If they get it wrong, things can go south fast. Miss the mark and underestimate demand? That leads to shortages, delayed treatments, and a lot of stress for patients. Overshoot, and you're left with piles of unused meds and financial headaches. This is especially risky for asthma treatments, since people rely on their meds to breathe, stay out of the hospital, and just get through daily life.

One of the trickiest parts of forecasting for asthma meds is dealing with seasonality and environmental effects. Asthma flares up at certain times of the year—like in the colder months when people catch more respiratory infections. Bad air days, high pollen, and humidity also push up demand. If there's a spike in air pollution or pollen, you can bet more people will reach for their inhalers.

To really see these trends, you need good data on both health and the environment. Public datasets are a goldmine for this. You can look at national health stats for hospital admissions, pharmacy sales records showing how much medication gets used, or environmental data tracking air quality and pollution.

Hospital admission numbers highlight when asthma worsens and when people need more medication. Pharmacy sales show how usage changes over time. Environmental data ties in outside factors like weather and pollution that drive up demand. By putting all this together, researchers get a clearer view of what drives the asthma medication market and what influences it.

This study dives into public data to look at the asthma medication market in Spain and Europe. The goal is to spot patterns in sales and seasonal health data, then build a model to predict how much medication people will need. This model helps pharmaceutical companies make smarter decisions about what to produce and where to send it.

Using public data matters here—there's no access to internal company data. Public sources are reliable, transparent, and perfect for academic research. Plus, they let you compare different regions and time periods, so you can spot big-picture trends in how asthma gets treated and how medication use changes across Europe.

Methodology

We're taking a hands-on, step-by-step approach to build a demand forecasting model for asthma medications. Since Chiesi didn't provide internal data, we're sticking to public datasets. Here's how it'll go: data collection, data prep, model selection, training and testing, evaluation, and finally, visualization. I'll break down every part so it's clear what's happening and why.

First up is data collection. To track how demand for asthma meds changes over time, we need two main data sources. For sales, we're pulling info from Datos.gob.es, Spain's national open-data portal. They break down medicine use by therapeutic group, so we can zoom in on respiratory drugs. We'll also look at IQVIA MIDAS Global Data, which sums up pharmaceutical sales across Europe, both in stores and hospitals. But sales numbers alone don't tell the whole story. We also need to factor in outside influences that drive up demand, like air pollution or flu season. For that, we'll use hospital admissions from Spain's National Hospital Discharge Database (CMBD), pollution stats from the European Environment Agency, and Eurostat's data on respiratory diseases. These sources help us spot seasonal peaks and understand what's driving them.

Once we've got the data, it needs cleaning and organizing. That means tossing out missing or irrelevant info and lining everything up so it's easy to work with—by month, by quarter, or whatever makes sense for seasonal patterns. We'll make sure all the data covers the same time periods and regions so we can actually compare things. Sometimes we'll create new variables, like the average pollution level each month or a flag for flu season, to help the models pick up on what really matters. Getting this part right is key—if the data's a mess, the model can't learn anything useful.

Next is picking the right models. We'll try out two main types: classic time series models (like ARIMA and SARIMA) and machine learning models (like Random Forest and XGBoost). Time series models do a good job spotting trends and seasonality in past data. Machine

learning models, on the other hand, are more flexible and can juggle lots of factors at once—think pollution, hospital admissions, and so on. By putting both types to the test, we'll see which one nails the forecasts.

After that, it's time to train and test. We'll split the data in two—one chunk for training, one for testing. The idea is simple: teach the model on one set, then see if it can predict demand on new data. This is where we find out if the model actually works, not just on stuff it's already seen, but on fresh info too.

The whole point of this study is to build a model that predicts demand for asthma meds better than the usual methods. By using public data and controlling for things like season, pollution, and spikes in respiratory illness, we get a much clearer picture of what's driving demand. We expect to find that people need more asthma medication during certain times of year—cold seasons or stretches when air quality tanks.

We also think the machine learning models, especially Random Forest and XGBoost, will beat the time series ones when we throw lots of predictors into the mix. They should pick up on complicated links between sales, hospital stays, and environmental conditions, making their predictions more accurate. Meanwhile, time series models will still be handy for spotting big-picture trends and cycles over time.

Conclusion

We'll use MAE, RMSE, and MAPE to see which model actually works best. The one with the lowest error rates and solid results with new data—yeah, that's the winner. If we can nail that, a pharmaceutical company could use the model to plan production and distribution, even if they don't have all their internal numbers on hand.

Besides the model itself, I'll lay out the results with graphs and charts that show how demand for asthma meds shifts over time, and what outside factors make those numbers jump or drop. These visuals should make the story behind the data clear, so people in the industry can actually use the findings to make decisions.

Big picture? This research proves you can take public data and build real forecasting tools that help companies like Chiesi—and others in health—plan their supply chains, cut down on waste, and, most importantly, make sure patients get their meds when they need them.

Bibliography

Fourkiotis, K. & Tsadiras, A., 2024. Applying Machine Learning and Statistical Forecasting Methods for Enhancing Pharmaceutical Sales Predictions. *Forecasting*, 6(1), pp.123–140. Available at: <https://www.mdpi.com/2571-9394/6/1/10>

Rathipriya, R., Sridevi, S. & Rajalakshmi, P., 2023. Demand forecasting model for time-series pharmaceutical data using shallow and deep neural networks. *Neural Computing and Applications*. Available at: <https://link.springer.com/article/10.1007/s00521-022-07889-9>

Ripatti, V., 2025. Forecasting Prescription Medication Utilization: SARIMA, Prophet, XGBoost and LSTM Models. MSc Thesis, Aalto University. Available at: <https://aaltodoc.aalto.fi/bitstreams/17e24b2b-946a-4013-8eb8-999fa010f120/download>

Majumdar, A., Singh, R. & Kumar, S., 2025. AI-Driven Optimization of Pharmaceutical Supply Chains. In: *ICDAM 2024*. Springer, pp. 215–228. Available at: https://link.springer.com/chapter/10.1007/978-3-032-03072-6_18

Angula, M. & Dongo, T., 2024. Impact of AI and ML on Forecasting Medication Demand in Public Pharmaceutical Systems. *GSC Biological and Pharmaceutical Sciences*, 7(3), pp.45–52. Available at: <https://gsconlinepress.com/journals/gscbps/sites/default/files/GSCBPS-2024-0071.pdf>

Santilli, P., 2024. Pharmaceutical Forecasting: Leveraging Intelligence for Accurate Demand Prediction. *SCIP Insights*. Available at: <https://www.scip.org/news/683019/Pharmaceutical-Forecasting-Leveraging-Intelligence-for-Accurate-Demand-Prediction-.htm>

Ross, T., 2024. Approaches to Pharmaceutical Demand Forecasting. *Mirador Global White Paper*. Available at: <https://miradorglobal.com/wp-content/uploads/2024/04/Approaches-to-Pharmaceutical-Demand-and-Forecasting.pdf>

Rizaldy, R., Sari, D. & Prasetyo, A., 2024. Comparative Analysis of Demand Forecasting Methods in PharmaHealth Group. *International Journal of Current Science Research and Review*, 7(8), pp.1208–1215. Available at: <https://ijcsrr.org/wp-content/uploads/2024/08/40-1208-2024.pdf>

Tonelli, M., 2021. Demand Forecasting in Pharmaceutical Supply Chains: Novo Nordisk Case Study. MSc Thesis, Universitat de Barcelona. Available at: https://diposit.ub.edu/dspace/bitstream/2445/186010/2/tfm_tonelli_mattia.pdf

Kwarteng, E. & Andreevich, A., 2024. Comparative Analysis of ARIMA, SARIMA and Prophet Model in Forecasting Antidiabetic Drug Demand. *Research & Development*, 5(4),

pp.112–120. Available at:

<https://sciencepublishinggroup.com/article/10.11648/j.rd.20240504.13>